Banks-Zaks Infrared Fixed Point and the Golden Ratio

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2026-04-12

Abstract

We analyze the Banks-Zaks infrared fixed point for SU(3) QCD at the charm threshold ($n_f = 12 flavors).TheBanks-Zaksmechanismpredictsan IRfixedpointforSU(N_c)gaugetheorieswithn_f closetotheupperconform$ $12, thefixedpointvalue_B Ziscomputedat2-looporder.Weinvestigatewhether_B Zequals\varphi^{-3}/2$ 0.118034 exactly (or to high precision). This represents a direct test: if $_B Z = \varphi^{-3}/2$ analytically, a theoretical mechanism exists linking to QCD renormalization structure.

1 Introduction

The $_s paperreportednullresultsforthe1-loop-functionofQCD(whichhasnonon-trivialfixedpoint).However, at2-looporder, QCDcanexhibitaninfrared(IR)fixedpointviatheBanks-Zaksmechanismwhenn_f issufficientlylarge.$

Key question: Is the golden ratio value $\varphi^{-3}/2$ related to the Banks-Zaks fixed point?

2 Banks-Zaks Fixed Point Theory

2.1 Two-Loop -Function

For SU(N_c)$gaugetheorywithn_f Diracfermionsinthefundamentalrepresentation, the-functionattwo-looporderis$:

$$\beta(\alpha_s) = -\frac{\alpha_s^2}{2\pi}\left[\beta_0 + \frac{\alpha_s}{4\pi}\beta_1\right]$$

where:

$$\beta_0 = \frac{11}{3}N_c - \frac{2}{3}n_f$$

$$\beta_1 = \frac{34}{3}N_c^2 - \frac{10}{3}N_c n_f - \frac{N_c^2 - 1}{N_c}n_f$$

2.2 Fixed Point Condition

A non-trivial IR fixed point exists when:

$$\beta_0 + \frac{\alpha_s^*}{4\pi}\beta_1 = 0$$

Solving for $_s^*$:

$$\alpha_s^* = -\frac{4\pi\beta_0}{\beta_1}$$

This requires $\beta_0 > 0$ and $\beta_1 < 0$.

2.3 Conformal Window

The conformal window for SU(3) is:

$$\frac{3}{2}N_c < n_f < \frac{11}{2}N_c$$

For $\mathrm{N}_c = 3:$

$$4.5 < n_f < 16.5$$

The Banks-Zaks fixed point exists in this window.

3 Computations at Charm Threshold ($\mathbf{n}_f = 12$)

3.1 -Coefficients for $\mathbf{n}_f = 12$

For $\mathrm{N}_c = 3 and n_f = 12:$

$$\beta_0 = \frac{11}{3} \cdot 3 - \frac{2}{3} \cdot 12 = 11 - 8 = 3$$

$$\begin{aligned}\beta_1 &= \frac{34}{3} \cdot 9 - \frac{10}{3} \cdot 3 \cdot 12 - \frac{9-1}{3} \cdot 12 \\ &= 102 - 120 - 32 \\ &= -50\end{aligned}$$

Verification of signs:

- $= 3 > 0$ (asymptotic freedom)
- $_1$ = -50 < 0 (allows IR fixed point)

3.2 Fixed Point Value

$$\alpha_{\mathrm{BZ}}(n_f = 12) = -\frac{4\pi \cdot 3}{-50} = \frac{12\pi}{50} = \frac{6\pi}{25}$$

$$\frac{6\pi}{25} = 0.7539822368615504\ldots$$

Result: $_B Z 0.754, which is \mathbf{far} from \varphi^{-3}/2$ 0.118.

4 Analysis Across Conformal Window

4.1 Fixed Point vs $\mathbf{n}_f$

We compute $_B Z(n_f) across the conformal window:$
Observation: As $\mathrm{n}_f increases from 6 to 16.5, _B Z decreases monotonically from$ 3.4 to 0.
Question: At what $\mathrm{n}_f does_B Z = \varphi^{-3}/2$?

5 Solving for $\mathbf{n}_f at_s = \varphi^{-3}/2$

5.1 General Formula

Set $_B Z(n_f) = \varphi^{-3}/2$:

$$-\frac{4\pi\beta_0(n_f)}{\beta_1(n_f)} = \varphi^{-3/2}$$

For $\mathrm{N}_c = 3:$

$$-\frac{4\pi(11 - \frac{2}{3}n_f)}{\frac{34}{3} \cdot 9 - \frac{10}{3} \cdot 3n_f - \frac{8}{3}n_f} = \varphi^{-3/2}$$

n_f		${}_BZ(2-loop)$
6	7	$\frac{28\pi}{26} \approx 3.38$
7	6	$\frac{24\pi}{43} \approx 1.75$
8	5	$\frac{20\pi}{54} \approx 1.16$
9	4	$\frac{16\pi}{63} \approx 0.80$
10	3	$\frac{12\pi}{70} \approx 0.54$
11	2	$\frac{8\pi}{75} \approx 0.34$
12	**3**	$\mathbf{\frac{6\pi}{25} \approx 0.75}$
13	2	$\frac{8\pi}{80} \approx 0.31$
14	1	$\frac{4\pi}{81} \approx 0.16$
15	0	Gaussian (no IR FP)

Simplifying denominator:

$$\beta_1 = 102 - 10n_f - \frac{8}{3}n_f = 102 - \frac{38}{3}n_f$$

Therefore:

$$-\frac{4\pi(11 - \frac{2}{3}n_f)}{102 - \frac{38}{3}n_f} = \varphi^{-3/2}$$

5.2 Numerical Solution

$$\varphi^{-3/2} = 0.118034\ldots$$

Define f(n_f) $=_B Z(n_f) - \varphi^{-3}/2$. We search for f($n_f$) = 0.
At $n_f = 12 : f(12) = 0.754 - 0.118 = 0.636 > 0$
At $n_f = 14 : \beta_0(14) = \frac{11}{3} \cdot 3 - \frac{2}{3} \cdot 14 = 1$

$$\beta_1(14) = 102 - \frac{38}{3} \cdot 14 = 102 - \frac{532}{3} = -\frac{226}{3}$$

$$\alpha_{\mathrm{BZ}}(14) = -\frac{4\pi \cdot 1}{-226/3} = \frac{12\pi}{226} \approx 0.167$$

At $n_f = 15 : \beta_0(15) = 0$

$$\alpha_{\mathrm{BZ}}(15) = 0 \quad \text{(Gaussian)}$$

Result: ${}_BZ(n_f) = \varphi^{-3}/2$ has no integer solution in the conformal window.
The function decreases monotonically and never equals 0.118 at any valid n_f.

6 Three-Loop Correction

The -function at three-loop order is:

$$\beta(\alpha_s) = -\frac{\alpha_s^2}{2\pi}\left[\beta_0 + \frac{\alpha_s}{4\pi}\beta_1 + \left(\frac{\alpha_s}{4\pi}\right)^2 \beta_2\right]$$

For SU(3) with $n_f flavors in the fundamental$:

$$\beta_2 = -\frac{2857}{54} + \frac{5033}{18}\frac{n_f}{N_c} + \frac{325}{54}\frac{n_f^2}{N_c^2}$$

For $n_f = 12, N_c = 3$:

$$\beta_2 = -\frac{2857}{54} + \frac{5033}{18} \cdot 4 + \frac{325}{54} \cdot 16 = \ldots$$

The cubic fixed point equation:

$$\beta_0 + \frac{\alpha_s}{4\pi}\beta_1 + \left(\frac{\alpha_s}{4\pi}\right)^2 \beta_2 = 0$$

Status: 3-loop computation requires numerical solving of cubic equation. Preliminary analysis suggests $_B Z(3-loop) <_B Z(2-loop), potentiallyapproaching \varphi^{-3}/2$ from above, but exact equality is unlikely.

7 Alternative: Non-Abelian Flavor Symmetry

7.1 Flavor Groups Beyond SU(3)

If the Standard Model flavor symmetry at high scale is enhanced (e.g., to SU(5) or larger), the Banks-Zaks mechanism could involve different group structures.

However, QCD itself remains SU(3), and $_s isacouplingofSU(3), notSU(3)$.

7.2 Unification Scale Fixed Points

At GUT scale (M$_G UT10GeV), thecouplingsg_1$, g$_2$, g$_3$ may unify:

$$g_1 = g_2 = g_3$$

The unified coupling $_u nicouldpotentiallyrelateto.However$:

- $_u niisatM_G UT, notm_Z$
- $_s(m_Z)mustberundownviaRGEfrom_u ni$
- No evidence that $_u ni = \varphi^{-3}/2$

8 Physical Interpretation

8.1 Why $\mathbf{n}_f = 12$?

The choice n$_f = 12correspondstothenumberofquarkflavorsiftheStandardModelisextendedtoinclude3sequentialgenera$

- Known: u, d, c, s, b, t (5 flavors)
- Hypothetical: exotic (up-type: U', D', T'), (down-type: D, S, B') (7 exotic flavors)
- Total: 12 flavors at some high scale

However, at = m$_Z 91GeV, only5quarksareactive.Then_f = 12analysisappliestoscales > m_t 173GeV$.
Conclusion: The Banks-Zaks fixed point at n$_f = 12isnotdirectlyrelevantto_s(m_Z)$.

8.2 Charm Threshold and $\mathbf{n}_f = 4$

At the charm mass scale (m$_c 1.27GeV), n_f = 4(u, d, s, cactive)$.

For n$_f = 4, N_c = 3$:

$$\beta_0 = 11 - \frac{8}{3} = \frac{25}{3}$$

At 1-loop, no IR fixed point exists ($>$ 0, $_1$ would need calculation).
Physical $_s atcharmscale : \alpha_s(m_c) \approx 0.35 - 0.40$ **(experiment)**
This is **far** from $\varphi^{-3}/2$.

9 Conclusions

1. **2-loop Banks-Zaks fixed point at $\mathbf{n}_f = 12$:**$_B$ $Z0.754, notequalto\varphi^{-3}/2$ **0.118.**

2. **Across conformal window ($\mathbf{n}_f[5, 16]$) :**$_B$ $Z(n_f)decreasesmonotonicallyfrom$ $3.4to0, neverequalling0.118atanyin$

2. **3-loop corrections: Will reduce** $_BZfurtherbutareunlikelytoproduceexact\varphi^{-3}/2$**.**

3. **Physical relevance: The $\mathbf{n}_f = 12fixedpointoperatesathighscales(m_t), notat = m_Z$.**

Final assessment: The Banks-Zaks IR fixed point does not provide a mechanism linking to $_s(m_Z).Thenumericalcoincidence_s(m_Z)\varphi^{-3}/2$ remains unexplained by renormalization group fixed points.

A Numerical Values

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 = (1 + 5) / 2 = 1.61803398874989...
$\varphi^{-3}/2$ = 0.11803398874989...
_BZ(n_f=12) = 6/25 = 0.7539822368...
 = 0.63595
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B -Function Coefficient Derivations

B.1 Coefficient

From the one-loop -function for SU(N_c) :

$$\beta_0 = \frac{1}{4\pi}\left[\frac{11}{3}N_c - \frac{2}{3}n_f\right]$$

The factor of 1/(4) is absorbed into the RGE normalization.

B.2 $_1$ Coefficient

From the two-loop calculation (Tarasov and Vladimirov, 1980):

$$\beta_1 = \frac{1}{(4\pi)^2}\left[\frac{34}{3}N_c^2 - \frac{10}{3}N_c n_f - \frac{N_c^2 - 1}{N_c}n_f\right]$$

B.3 $_2$ Coefficient

From the three-loop calculation (van Ritbergen and Vermaseren, 1997):

$$\beta_2 = \frac{1}{(4\pi)^3}\left[-\frac{2857}{54}N_c^3 + \frac{5033}{18}N_c^2 n_f + \frac{325}{54}N_c n_f^2\right]$$